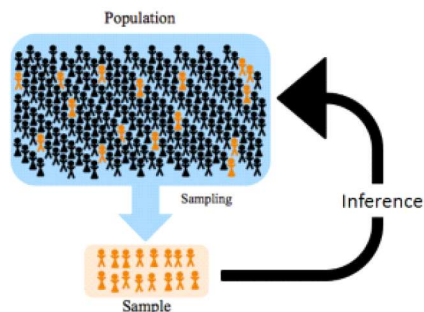


Introduction

02 September 2025 07:20

Statistical science has three primary aspects:

- Design "How" to collect data
e.g. Marketing offers on pizza w/ cheese or without
Causal conclusions
- Description "Summarizing" the data
- Inference Evaluations such as estimation & prediction
Account for variability from samples to population.



Link: Penn State

Although we obtain data for the sample subjects, our ultimate interest is on the population from which the sample is taken.

Population and sample

The **population** is the total set of subjects of interest. A **sample** is the set of subjects from the population for which data are available.

sample **Statistic** is used to estimate the population **parameters**.

A statistic is any quantity computed from values in a sample

If we obtain a random sample and calculate a sample statistic from that sample, the sample statistic is a random variable

That means there is $E[S]$, $V[S]$, $dist(S)$

Population parameters are fixed.

SAMPLING DISTRIBUTION
of STATISTIC

4.1 (Sampling Distribution)

The sampling distribution of a statistic is a probability distribution based on a large number of samples of size n from a given population.

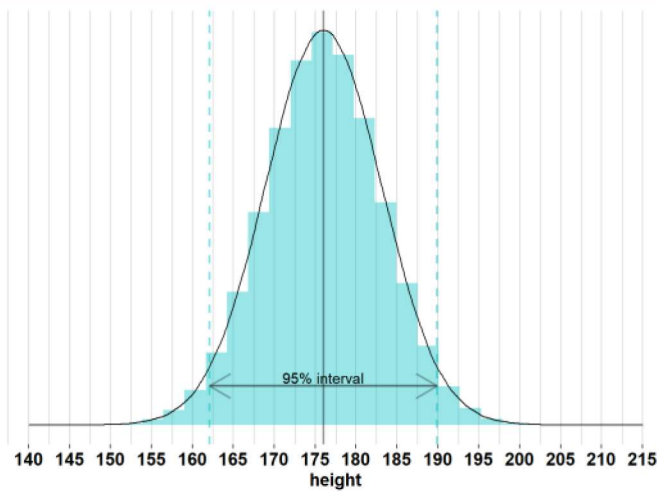
Suppose the interest is in estimating Avg. Height of Campus Residents

We take a sample of size 50, compute mean, call it $\bar{x}_1$

" " second " " " " " " " " $\bar{x}_2$

" " k^{th} " " " " " " " " $\bar{x}_k$

hist($\bar{x}_i$)



Let $X_1, X_2, \dots, X_n$ be n independent random variables, each having the same probability distribution $f(x)$. Define $X_1, X_2, \dots, X_n$ to be a **random sample** of size n from the population $f(x)$ and write its joint probability distribution as

$$f(x_1, x_2, \dots, x_n) = f(x_1)f(x_2) \cdots f(x_n).$$

Example 4.1 (Sample Means with a Small Population: Pumpkin Weights)



<https://online.stat.psu.edu/stat500/Lesson04>

In this example, the population is the weight of six pumpkins (in pounds) displayed in a carnival "guess the weight" game booth. You are asked to guess the average weight of the six pumpkins by taking a random sample without replacement from the population.

Pumpkin	A	B	C	D	E	F
Weight (in pounds)	19	14	15	9	10	17

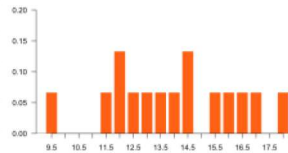
Since we know the weights from the population, we can find the population mean.

$$\mu = \frac{19 + 14 + 15 + 9 + 10 + 17}{6} = 14 \text{ pounds}$$

Take random samples of size 2

A, B:
A, C:
|
E, F:

$$\bar{x} = \frac{19+14}{2} = 16.5$$



$E(\bar{x})$

$\bar{x}$	9.5	11.5	12.0	12.5	13.0	13.5	14.0	14.5	15.5	16.0	16.5	17.0	18.0
Probability	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{1}{15}$

$$P(\text{sample mean} = \text{pop mean}) = \text{small} = 0.067$$

The mean of the sample means is

$$\begin{aligned} \mu_{\bar{x}} &= \sum \bar{x}_i f(\bar{x}_i) = 9.5 \left(\frac{1}{15} \right) + 11.5 \left(\frac{1}{15} \right) + 12 \left(\frac{2}{15} \right) \\ &+ 12.5 \left(\frac{1}{15} \right) + 13 \left(\frac{1}{15} \right) + 13.5 \left(\frac{1}{15} \right) + 14 \left(\frac{1}{15} \right) \\ &+ 14.5 \left(\frac{2}{15} \right) + 15.5 \left(\frac{1}{15} \right) + 16 \left(\frac{1}{15} \right) + 16.5 \left(\frac{1}{15} \right) \\ &+ 17 \left(\frac{1}{15} \right) + 18 \left(\frac{1}{15} \right) = 14 \end{aligned}$$

sample mean

$$E[\bar{x}] = \mu$$

do it with $n=5$

Now, let's do the same thing as above but with sample size $n=5$.

Sample	Weights	$\bar{x}$	Probability
A, B, C, D, E	19, 14, 15, 9, 10	13.4	1/6
A, B, C, D, F	19, 14, 15, 9, 17	14.8	1/6
A, B, C, E, F	19, 14, 15, 10, 17	15.0	1/6
A, B, D, E, F	19, 14, 9, 10, 17	13.8	1/6
A, C, D, E, F	19, 15, 9, 10, 17	14.0	1/6
B, C, D, E, F	14, 15, 9, 10, 17	13.0	1/6

$$= 0.167$$

The sampling distribution is:

$\bar{x}$	13.0	13.4	13.8	14.0	14.8	15.0
Probability	1/6	1/6	1/6	1/6	1/6	1/6

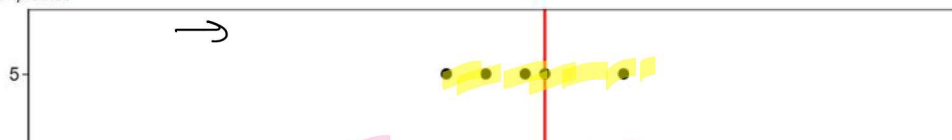
The mean of the sample means is...

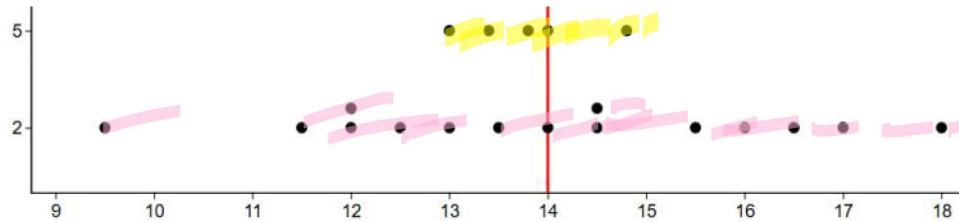
$$\mu = \left(\frac{1}{6} \right) (13 + 13.4 + 13.8 + 14.0 + 14.8 + 15.0) = 14 \text{ pounds}$$

The following dot plots show the distribution of the sample means corresponding to sample sizes of $n=2$ and of $n=5$.

Population Mean

Sample Size





Again, we see that using the sample mean to estimate population mean involves sampling error. However, the error with a sample of size $n = 5$ is on the average smaller than with a sample of size $n = 2$.

Sampling Error and Size

4.2 (Sampling Error)

The error resulting from using a sample characteristic to estimate a population characteristic

Sample size and sampling error: As the dotplots above show, the possible sample means cluster more closely around the population mean as the sample size increases. Thus, the possible sampling error decreases as sample size increases.

If the underlying population is normal, that is, $X_i \sim N(\mu, \sigma^2)$
then the dist of $\bar{X} \sim N(\mu, \sigma^2/n)$

For any random sample s.t. $E(X_i) = \mu$, $V(X_i) = \sigma^2$.
Then, $E(\bar{X}) = \mu$ & $V(\bar{X}) = \frac{\sigma^2}{n}$

Central Limit Theorem (Most fundamental result in statistics)

For a large sample size (we will explain this later), $\bar{x}$ is approximately normally distributed, regardless of the distribution of the population one samples from. If the population has mean μ and standard deviation σ , then $\bar{x}$ has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.
popⁿ with finite variance

- If the population is skewed and sample size small, then the sample mean won't be normal.
- When doing a simulation, one replicates the process many times. Using 10,000 replications is a good idea.
- If the population is normal, then the distribution of sample mean looks normal even if $n = 2$. Note the app in the video used capital N for the sample size.
- If the population is skewed, then the distribution of sample mean looks more and more normal when n gets larger.
- Note that in all cases, the mean of the sample mean is close to the population mean and the standard error of the sample mean is close to $\frac{\sigma}{\sqrt{n}}$.

https://online.statbook.com/stat_sim/sampling_dist/index.html

← Play with this tool

Ex 1

The weights of baby giraffes are known to have a mean of 125 pounds and a standard deviation of 15 pounds.

If we obtained a random sample of 40 baby giraffes,

- what is the probability that the sample mean will be between 120 and 130 pounds?
- what is the 75th percentile of the sample means of size $n = 40$?

$$P(-2.108 < Z < 2.108) = 0.96; \quad P(Z < 0.6744) = 0.75$$

$\mu = 125$, $\sigma = 15$, $n = 40$

a) $P\left(\frac{120-125}{15/\sqrt{40}} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < \frac{130-125}{15/\sqrt{40}}\right) = P(-2.108 < Z < 2.108) = 0.96$

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

b) $P(Z < z_\alpha) = 0.75$

$$z_\alpha \frac{\sigma}{\sqrt{n}} + \mu = 0.6744 \times \frac{15}{\sqrt{40}} + 125$$

$$= 126.56 \text{ pounds}$$

$P(Z < z_\alpha) = 1 - \alpha$



4.2.2: Sampling Distribution of the Sample Proportion

The distribution of the sample proportion approximates a normal distribution under the following 2 conditions.

Over the years the values of the conditions have changed. The examples that follow in the remaining lessons will use the first set of conditions at 5, however, you may come across other books or software that may use 10 or 15 for this value.

Book (Minitab)

1. $np \geq 5$
2. $n(1-p) \geq 5$

1990-2000s

1. $np \geq 10$
2. $n(1-p) \geq 10$

Current

1. $np \geq 15$
2. $n(1-p) \geq 15$

4.5 (Sampling Distribution of the Sample Proportion)

If any set of the two conditions listed above are satisfied, the **sampling distribution of the sample proportion** is...

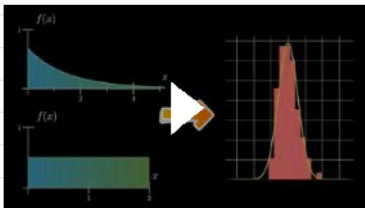
- approximately normal with mean, $\mu = p$
- standard deviation [standard error], $\sigma = \sqrt{\frac{p(1-p)}{n}}$

If the sampling distribution of $\hat{p}$ is approximately normal, we can convert a sample proportion to a z-score using the following formula:

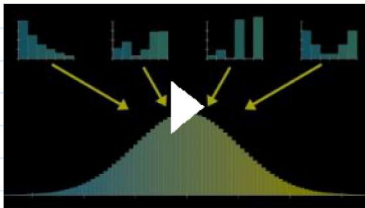
$$z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

We can apply this theory to find probabilities involving sample proportions.

[The Central Limit Theorem Clearly Explained!](#)



Watch this cool video: [But what is the Central Limit Theorem?](#)



→ Mean ↙ ↘
→ Proportion
→ Variance